BT5340 Assignment 3 Report

# Protein Folding as a Nonlinear Dynamical System

## Attractor Basin Geometry, Hydrophobic Collapse, and Empirical Signatures in Protein Language Model Embedding Space

Team Perutz

Karthik, Ruhaib, Saahas

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# Abstract

Protein folding has traditionally been described through thermodynamic energy landscape and kinetic frameworks, where the native state corresponds to a free-energy minimum at the bottom of a funnel-shaped landscape. This project proposes a complementary nonlinear dynamical systems (NLD) perspective, modelling the folding process as the evolution of a conformational state toward an attractor in a high-dimensional phase space. Because mapping full folding trajectories is computationally infeasible and considered a non trivial problem and too ambitious to take up within the scope of a course project, we leverage protein language model (pLM) embedding space as a proxy for the folding landscape.

Using the models ESM-2 (650M parameters) for structural embeddings and ESM-1v for masked-marginal stability scoring, we systematically introduced all 4,674 single-point mutations across three well-characterised model proteins:ubiquitin (76 aa), barnase (110 aa), and GB1 (56 aa) and analysed the geometric structure of the resulting embedding perturbations. Our results show that:

1. L2 embedding distance from the wild-type correlates positively and significantly with predicted destabilisation (Spearman $r = 0.17$–$0.57$, all $p < 10^{-10}$; $R^2 = 0.03$–$0.22$), with the relationship being monotonic but nonlinear which is consistent with a attractor basin geometry;
2. Funnel scores exceed the permutation null by 75–1,445 standard deviations which provides strong evidence for attractor like basin geometry
3. Displacement vectors are anisotropic, with destabilising mutations preferentially pushing embeddings in shared directions away from the wild-type, while stabilising mutations show more uniform distribution;
4. ESM-2 attention-derived contact maps recover known topology-specific structural contacts without structural supervision, and perturbation sensitivity co-localizes with contact-forming positions, directly linking attractor basin geometry to the hydrophobic interactions that drive folding.

Together, these findings support the hypothesis that pLM embedding space encodes genuine biophysical knowledge consistent with attractor basin topology, and that pLMs acquire more than mere sequence statistics during their pretraining process.


# 1 Introduction

The protein folding problem tries to answer the question of how a linear amino-acid chain reaches its native three-dimensional structure, which to this day remains one of the central questions in molecular biology. The dominant theoretical framework describes folding through an energy landscape shaped like a funnel [1], where the native state occupies the global free-energy minimum and the folding trajectory descends through an ensemble of increasingly native-like conformations. The hydrophobic effect, where non-polar residues are driven to the protein core to minimize solvent-exposed surface area, has long been recognized as the dominant force shaping this landscape [2].

In this project, we propose an alternative yet complementary perspective rooted in **nonlinear dynamical systems (NLD) theory**. Rather than describing folding as descent on a free-energy surface, we model it as the evolution of a conformational state vector toward an *attractor* in a high-dimensional phase space. Under this framework, the native state is an attractor; the folding funnel corresponds to its basin of attraction; and the ongoing conformational fluctuations observed in folded proteins reflect the dynamics of a system near the fixed point of the system

While mapping the complete protein folding trajectory in an NLD framework is our ultimate interest, the sheer computational cost puts it beyond the scope of an undergraduate project. Therefore, we have designed a constrained study to test our primary hypothesis and establish a foundational empirical grounding within our available computing limits. To achieve this we leverage the embedding spaces of **protein language models (pLMs)** as computationally accessible proxies for the folding landscape. pLMs such as ESM-2 [3] are trained on vast datasets of evolutionary sequence corpora numbering in the millions, and they implicitly encode structural and thermodynamic viability in their learned representations, because the sequences that survive natural selection are precisely those that fold stably. However, these modesl are still essentially black boxes, as they are pretrained in an unsupervised manner (eg., fill in the missing residues or predict the next residue), without any explicit goal to learn biological knowlegde. Thus, there is still a lot of debate on whether these models and their learned representations actually learn meaningful biological knowledge.

One meaningful way of assessing this is to try and do an intrinsic evaluation of their embeddings in isolation so as to dissect what contexts they store, and whether it matches up with known biological sequence function and structure. The key insight is that if pLM embeddings do capture biophysical reality, then our hypothesis is that the wild-type embedding should sit at the centre of a basin and act as an attractor with the mutant embeddings displaced radially in proportion to their destabilising effect. This would also also support the notion that these pLMs do indeed learn biological information through their pretraining process.

We test this hypothesis by exhaustively generating all single-point mutations for three model proteins (ubiquitin, barnase, and GB1),These specific proteins serve as great structural benchmarks as they are well model systems in biophysics. Due to their small size, high stability, and diverse architectural topologies that are ranging from the fast, two-state folding of ubiquitin and the beta-hairpin configurations of GB1, to the multi-state folding pathways of barnase, they provide a large spectrum of folding behaviors. We do this by computing ESM-2 embeddings for each mutant, and analysing the geometry of the resulting displacement vectors through PCA, t-SNE, UMAP, funnel-score analysis, directionality testing, and attention derived contact maps. ESM-1v masked-marginal scores [4] serve as a zero-shot proxy for $\Delta\Delta G$ (destabilisation), which we validate against experimental data from the S2648 benchmark [5]

# 2 Methods

## 2.1 Model Proteins

Three well-studied proteins were selected to span a range of sizes and topologies: **ubiquitin** (76 residues, $\alpha/\beta$ roll), **barnase** (110 residues, $\alpha+\beta$ ribonuclease), and **GB1** (56 residues, $\beta$-grasp fold). For each protein, all 19 possible single-amino-acid substitutions were generated at every position, yielding 1,444, 2,166, and 1,064 mutants respectively (4,674 total).

Table 1: Mutant coverage per protein.

| Protein | Length | PDB | Mutants | Emb. Dim |
|---|---|---|---|---|
| Ubiquitin | 76 aa | 1UBQ | 1,444 | 1,280 |
| Barnase | 110 aa | 1BNI | 2,166 | 1,280 |
| GB1 | 56 aa | 1PGA | 1,064 | 1,280 |
| **Total** | | | **4,674** | |

## 2.2 ESM-2 Embeddings

ESM-2 (`facebook/esm2_t33_650M_UR50D`) was used. For each sequence, the model produces a per-residue hidden state of dimension 1,280. The mean-pooled embedding (excluding special `<CLS>` and `<EOS>` tokens) was taken as the sequence-level representation. The displacement vector for each mutant was computed as:

$$\mathbf{d} = \mathbf{e}_{\text{mut}} - \mathbf{e}_{\text{WT}}$$

and the L2 distance $\|\mathbf{d}\|_2$ and cosine distance were recorded.

## 2.3 ESM-1v Masked-Marginal Scoring

ESM-1v (`facebook/esm1v_t33_650M_UR90S_1`) was used for zero-shot stability prediction via masked-marginal scoring [4]. For each mutation at position $i$, the wild-type residue was replaced with the `[MASK]` token, and the log-likelihood ratio was computed:

$$\text{score} = -\left[\log P(\text{mut} \mid \text{mask}) - \log P(\text{wt} \mid \text{mask})\right]$$

The sign convention is such that positive values indicate destabilisation, aligning with the $\Delta\Delta G$ convention.

## 2.4 Experimental Validation Datasets

An experimental benchmark was used:

**S2648** [5]: 165 mutations across ubiquitin ($n$=11, 1UBQ), barnase ($n$=153, 1BNI), and GB1 ($n$=12, 1PGA) with measured $\Delta\Delta G$ values.

The S669 dataset was also examined but excluded because it uses UniProt accession IDs rather than PDB codes, making direct merging infeasible without external mapping.

### 2.5 Analysis Pipeline

The following analyses were performed on the 4,674 displacement vectors:

(a) Spearman correlation between L2 distance and ESM-1v stability scores;
(b) PCA of displacement vectors to visualise basin geometry;
(c) A funnel-score metric that captures how smooth and steep the embedding landscape is around the wild-type, tested against 1,000 randomly shuffled controls.
(d) A directionality test checking whether mutations push the embedding in shared preferred directions, or simply scatter randomly in all directions.
(e) Contact map extraction from APC-corrected ESM-2 attention weights;
(f) Perturbation sensitivity maps combining attention and positional L2;
(g) Positional sensitivity with Kyte–Doolittle hydrophobicity overlay [6];
(h) t-SNE at perplexities 15 and 30;
(i) UMAP at `n_neighbors` = 10, 15, and 30.

## 3 Results

### 3.1 Correlation: L2 Distance vs ESM-1v Stability Score

All three proteins show a positive and significant Spearman correlation between L2 embedding displacement from the wild-type and the ESM-1v destabilisation proxy. GB1 exhibits the strongest coupling ($r = 0.569$), consistent with its compact, well-packed $\beta$-grasp topology being especially sensitive to point mutations. The pooled correlation across all 4,674 mutants (using z-scored L2 to normalise across proteins) is $r = 0.202$ ($p = 3.2 \times 10^{-44}$). These results confirm the core hypothesis: mutants that are predicted to be more destabilising are embedded further away from the wild-type in ESM-2 space.

Table 2: Pearson $R^2$ for the linear component of the L2 distance ESM-1v destabilisation relationship. The lower $R^2$ relative to the Spearman $r$ values in Table 2 confirms that the embedding stability relationship is monotonic but nonlinear.

| **Protein** | $n$ | **Spearman** $r$ | $R^2$ | $p$**-value** |
|---|---|---|---|---|
| Ubiquitin | 1,444 | 0.173 | 0.028 | $3.6 \times 10^{-11}$ |
| Barnase | 2,166 | 0.276 | 0.064 | $3.3 \times 10^{-39}$ |
| GB1 | 1,064 | 0.569 | 0.216 | $2.8 \times 10^{-92}$ |

The Pearson $R^2$ values for the L2–ESM-1v relationship are lower than the squared Spearman $r$ , it would suggest under a linear assumption: $R^2 = 0.028$, 0.064, and 0.216 for ubiquitin, barnase, and GB1 respectively. This inconsistency in the values indicates that while the rank ordering of mutations by embedding distance is monotonically consistent with destabilization which was captured by Spearman $r$), the underlying relationship is nonlinear. The LOWESS curves in Figure 1 confirm this: Destabilization increases steeply with distance L2 at short distances, but saturates at larger displacements, consistent with a bounded attractor basin geometry in which extreme mutations all reach the basin boundary regardless of their destabilization magnitude. GB1's higher $R^2$ (0.216) relative to ubiquitin (0.028) and barnase (0.064) reflects a more tighter component. Linear component of the embedding stability suggests a strong relationship.

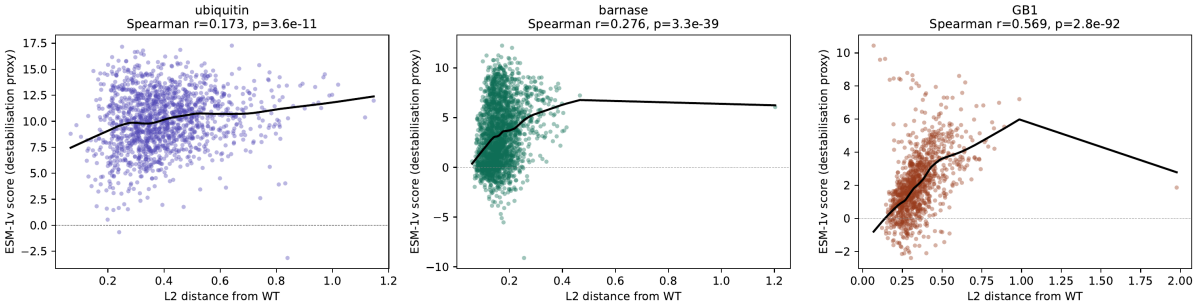



Figure 1: L2 distance vs ESM-1v score scatter plots with LOWESS fit for ubiquitin, barnase, and GB1.

## 3.2 Experimental Validation (S2648)

To assess whether the ESM-1v masked-marginal proxy captures real thermodynamic stability, we compared predicted scores against experimentally measured $\Delta\Delta G$ values from the S2648 benchmark.

GB1 shows strong agreement (Spearman $r = 0.790$, $p = 2.2 \times 10^{-3}$, $n$=12), validating the proxy for this protein.

Ubiquitin trends positive ($r = 0.518$) but is underpowered ($n$=11, $p = 0.10$).

Barnase produces a statistically significant negative correlation between the ESM-1v score and the experimental $\Delta\Delta G$ ($r = -0.258$, $p = 1.3 \times 10^{-3}$, $n = 153$). Further analysis reveals that this is not a simple ranking artefact: 88.9% of barnase entries in S2648 have negative experimental $\Delta\Delta G$ (stabilising), with a mean of $-1.36$ kcal/mol, confirming the extreme bias of the dataset toward the stabilising regime. Mainly, the anti-correlation persists within the stabilising subset alone ($r = -0.202$, $p = 0.018$, $n = 136$), demonstrating that ESM-1v assigns high destabilisation scores to experimentally stabilising barnase mutations rather than merely misranking them. This reflects a pretty important distinction between what ESM-1v measures and what calorimetry measures: the model could be encoding evolutionary plausibility (what natural selection tolerates), whereas the S2648 barnase entries represent engineered mutations that exceed the protein's evolutionary stability optimum. Barnase is under strong purifying selection for both catalytic activity and barstar-inhibitor compatibility; mutations that improve thermodynamic stability while disrupting these constraints are evolutionarily disfavoured and therefore flagged as anomalous by the model. The destabilising subset ($n = 17$) is underpowered for independent assessment ($p = 0.268$). This result does not undermine the attractor basin framework; barnase still achieves a funnel $z$-score of 541.2 and a significant L2-stability correlation ($r = 0.276$, $p < 10^{-38}$), but it reveals that the pLM attractor encodes evolutionary fitness constraints rather than pure thermodynamic stability, and that these two quantities diverge when mutations are engineered beyond natural sequence space.

Table 3: ESM-1v and L2 distance vs experimental $\Delta\Delta G$ (S2648 benchmark). $R^2$ values reflect the linear variance in experimental $\Delta\Delta G$ explained by each predictor independently.

| | | **ESM-1v vs exp $\Delta\Delta G$** | | | **L2 dist vs exp $\Delta\Delta G$** | | |
|---|---|---|---|---|---|---|---|
| **Protein** | $n$ | **Spearman $r$** | $R^2$ | $p$ | **Spearman $r$** | $R^2$ | $p$ |
| Ubiquitin | 11 | +0.518 | 0.133 | $1.02 \times 10^{-1}$ | +0.536 | 0.290 | $8.90 \times 10^{-2}$ |
| Barnase | 153 | −0.258 | 0.061 | $1.28 \times 10^{-3}$ | −0.217 | 0.063 | $7.01 \times 10^{-3}$ |
| GB1 | 12 | +0.790 | 0.623 | $2.22 \times 10^{-3}$ | +0.455 | 0.190 | $1.38 \times 10^{-1}$ |

Pearson $R^2$ values against experimental $\Delta\Delta G$ also show us an important distinction between ESM-1v and L2 as predictors. For GB1, ESM-1v achieves $R^2 = 0.623$ (62.3% of experimental variance explained), a strong result for a zero-shot predictor with no protein-specific training. By contrast, L2 distance alone achieves only $R^2 = 0.190$ for GB1 ($p = 0.138$, not significant), showing that embedding displacement is a noisier proxy for thermodynamic stability than the masked-marginal score itself. We think this is because ESM-1v encodes position-specific evolutionary information at each masked site, whereas L2 distance is a coarser global summary of how much the embedding moved. For ubiquitin, L2 achieves $R^2 = 0.290$ despite non-significance ($p = 0.089$, $n = 11$), which suggests that the relationship may be real but the sample is too small to confirm. Barnase $R^2$ values remain low for both predictors ($\leq 0.063$).

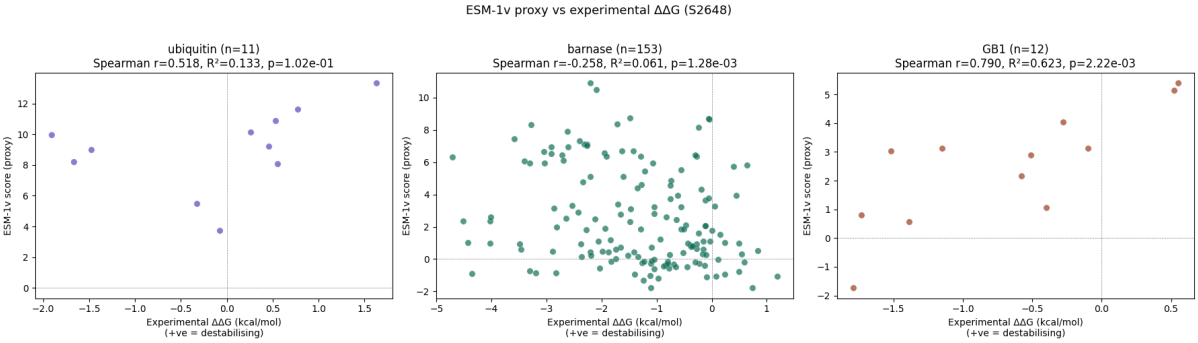



Figure 2: ESM-1v proxy score vs experimental $\Delta\Delta G$ from S2648 for ubiquitin, barnase, and GB1

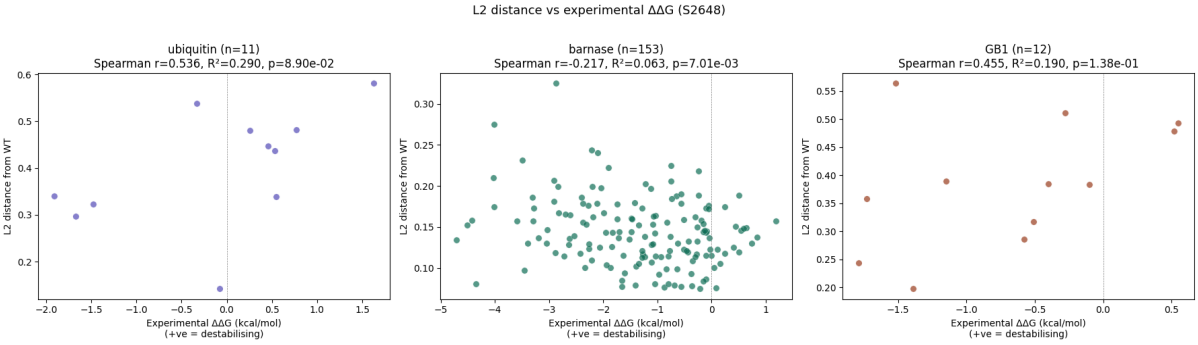



Figure 3: L2 distance from WT vs experimental $\Delta\Delta G$ from S2648 for ubiquitin, barnase, and GB1

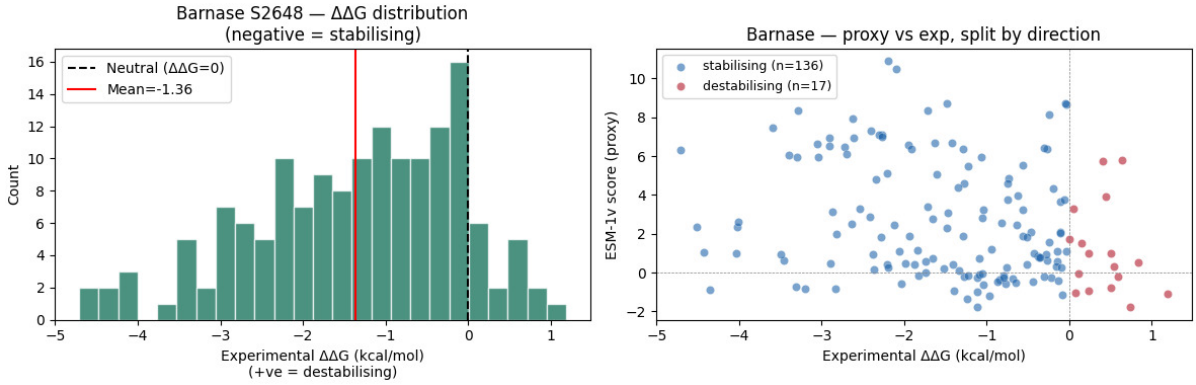



Figure 4: Diagnostic analysis of the barnase S2648 validation failure

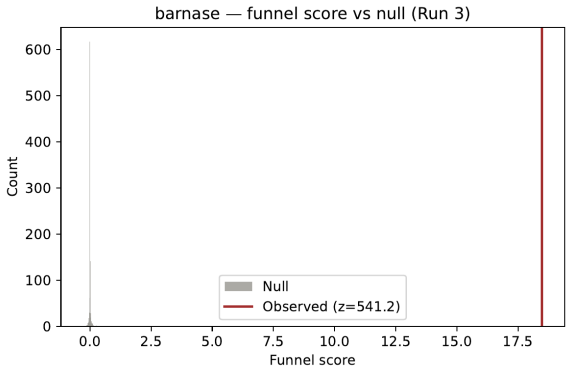
## 3.3 Funnel Score and Attractor Basin Test

The funnel score integrates three properties of the embedding landscape: directional correlation (Spearman $r$ between L2 distance and $\Delta\Delta G$ proxy), gradient magnitude (slope of the linear regression), and smoothness (inverse of within-shell variance normalised by gradient magnitude). A high funnel score indicates a smooth, monotonically increasing landscape emanating from the wild-type, which is exactly what our NLD attractor hypothesis predicts.

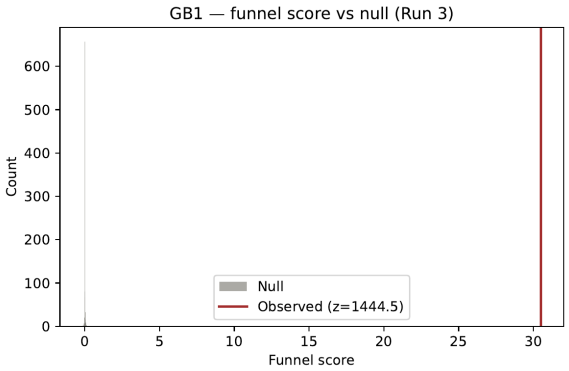
All three proteins show funnel scores exceeding the 1,000-permutation null distribution. The z-scores: 75.8 for ubiquitin, 541.2 for barnase, and 1,444.5 for GB1 represent extreme statistical significance ($p < 0.001$ in all cases). The observed funnel scores lie entirely outside the range of the null distributions, which cluster tightly around zero. This provides strong evidence that the embedding landscape is not random but ehxibits the structured, funnel-shaped geometry predicted by the attractor basin hypothesis.

Table 4: Funnel scores and permutation null comparison.

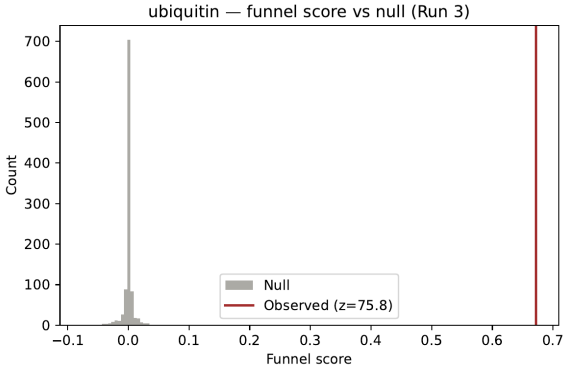
| **Protein** | **Funnel Score** | ***z*-score** | ***p* (perm)** |
|---|---|---|---|
| Ubiquitin | 0.672 | 75.8 | $< 0.001$ |
| Barnase | 18.47 | 541.2 | $< 0.001$ |
| GB1 | 30.50 | 1,444.5 | $< 0.001$ |




(a) Barnase ($z = 541.2$)




(b) GB1 ($z = 1{,}444.5$)




(c) Ubiquitin ($z = 75.8$)

Figure 5: Funnel score vs 1,000-permutation null distribution for each protein. The observed funnel score (red line) lies far outside the null distribution (grey histogram) in all cases.

## 3.4 PCA Basin Geometry

Principal component analysis of the displacement vectors yields a characteristic radial pattern that's consistent with the attractor basin model. In all PCA projections, the wild-type occupies the origin (marked with ★), and mutants spread outward in multiple directions. The coloration by the ESM-1v score shows that the most destabilizing mutations (red/warm) lie farthest from the origin, while neutral and stabilizing mutations (blue/cool) cluster near the center. Coloring by L2 distance confirms the same radial gradient! The top two principal components capture 12–18% of variance each, indicating that the displacement landscape is high-dimensional but has a significant low-rank structure aligned with stability.

Table 5: Variance explained by the top two PCA components.

| **Protein** | **PC1** | **PC2** |
|---|---|---|
| Ubiquitin | 17.9% | 13.9% |
| Barnase | 14.6% | 6.9% |
| GB1 | 17.6% | 11.8% |
| All proteins | 12.3% | 8.2% |

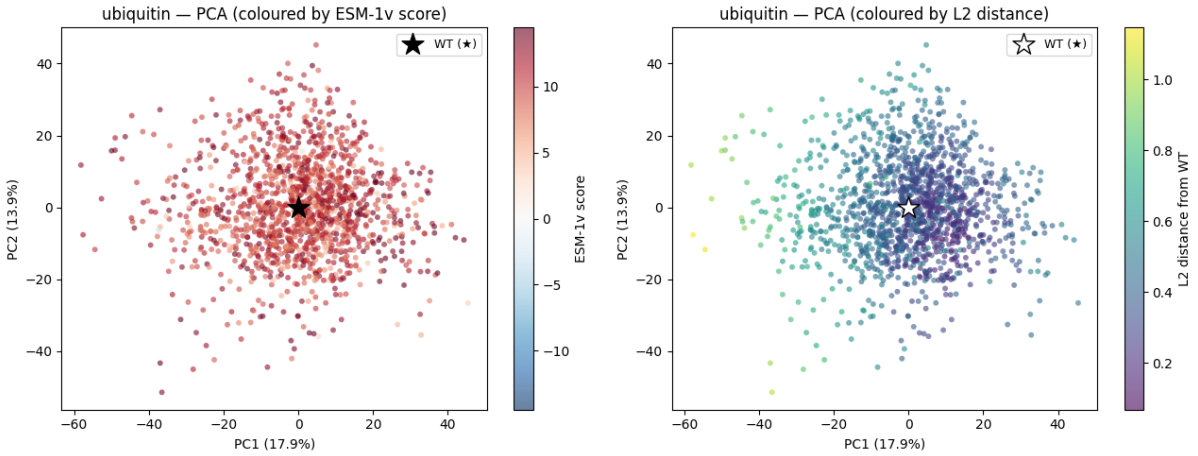



Figure 6: PCA displacement basin - Ubiquitin. Left: coloured by ESM-1v score. Right: coloured by L2 distance. The WT is marked with a star at the origin.

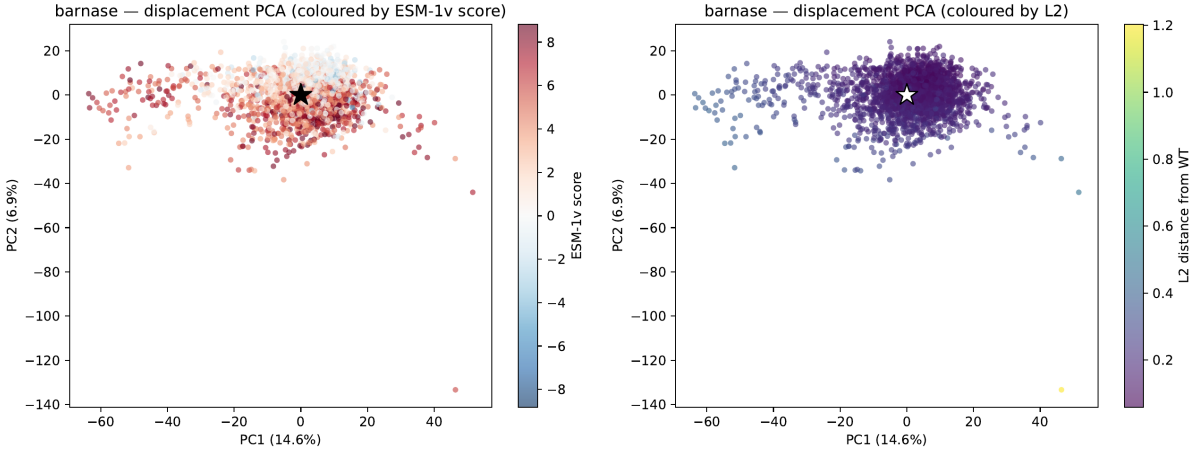



Figure 7: PCA displacement basin - Barnase.

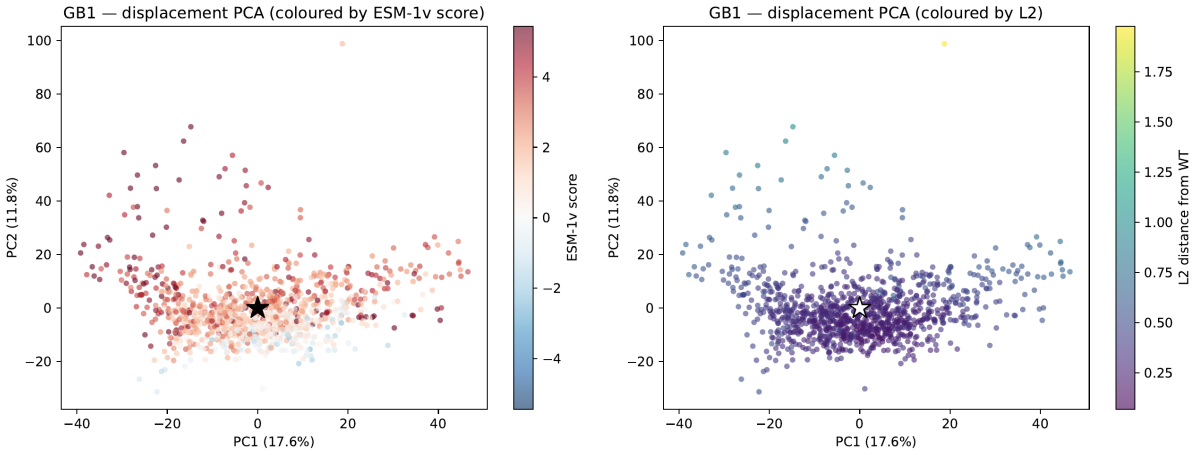



Figure 8: PCA displacement basin - GB1.

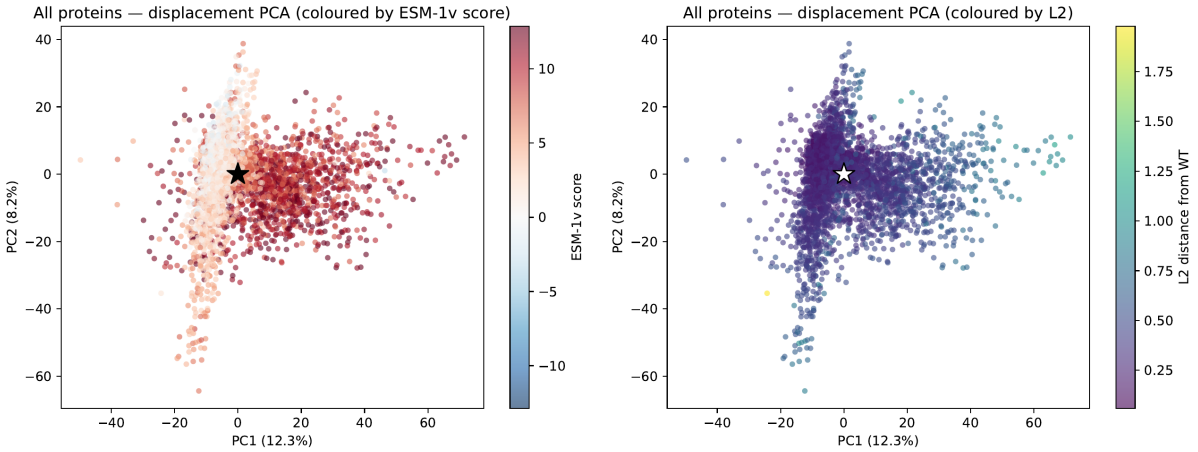



Figure 9: PCA displacement basin - all proteins combined.

## 3.5 Directionality and Anisotropy Analysis

If the embedding landscape were structurally featureless, mutation displacement vectors would point in random directions and the mean resultant length $R$ of the unit-normalised vectors would approach zero. Instead, the observed $R$ is significantly greater than the null expectation for all three proteins ($p < 0.001$, 1,000-permutation test), suggesting that mutations do not perturb the embedding isotropically.

Table 6: Isotropy test results per protein. $R_{\text{obs}}$ is the mean resultant length of unit-normalised displacement vectors; null statistics are derived from 1,000 permutations.

| **Protein** | $\boldsymbol{R}_{\text{obs}}$ | **Null** ($\boldsymbol{\mu} \pm \boldsymbol{\sigma}$) | $\boldsymbol{p}$ **(perm)** |
|---|---|---|---|
| Ubiquitin | 0.5563 | $0.0263 \pm 0.0005$ | $< 0.001$ |
| Barnase | 0.2949 | $0.0215 \pm 0.0004$ | $< 0.001$ |
| GB1 | 0.3404 | $0.0307 \pm 0.0006$ | $< 0.001$ |

Table 7: Mean resultant length $R$ classifies by stability class. Higher $R$ indicates greater directional coherence among displacement vectors within that class. Ubiquitin has no stabilising or neutral class because virtually all single-point mutants receive positive ESM-1v destabilisation scores, consistent with its unusually high intrinsic stability.

| **Protein** | **Stability class** | $n$ | $R$ |
|---|---|---|---|
| Ubiquitin | Mildly destabilising | 8 | 0.3621 |
| | Strongly destabilising | 1,434 | 0.5579 |
| Barnase | Stabilising | 190 | 0.1560 |
| | Neutral | 290 | 0.1715 |
| | Mildly destabilising | 388 | 0.2604 |
| | Strongly destabilising | 1,298 | 0.3919 |
| GB1 | Stabilising | 63 | 0.2419 |
| | Neutral | 132 | 0.1927 |
| | Mildly destabilising | 360 | 0.3275 |
| | Strongly destabilising | 509 | 0.4755 |

Further stratification by stability class reveals that *strongly destabilising* mutations show the highest directional coherence; they preferentialy push the embedding in shared directions (away from the attractor centre), while stabilising mutations exhibit more scattered directions. This directionality is consistent with the NLD hypothesis that the attractor has structured geometry with preferred escape routes.

## 3.6 Contact Map Analysis

Contact maps were extracted from ESM-2 attention weights using Average Product Correction (APC) to remove phylogenetic bias. Three panels were generated per protein:

1. **Predicted Contact Map:** APC-corrected attention-weight matrix, symmetrised and averaged across all layers and heads.
2. **Hydrophobic Interaction Matrix:** Outer product of Kyte–Doolittle hydrophobicity values [6], normalised to $[0, 1]$.
3. **Perturbation Sensitivity Map:** Element-wise product of the attention map with the average of positional mean-L2 values for each residue pair.

The predicted contact maps recover topology-specific structural features from attention weights alone, without structural supervision: ubiquitin shows subtle off-diagonal features consistent with its $\beta$-sheet strands, barnase shows elevated perturbation sensitivity near its C-terminal active site (residues ∼95–110), and GB1 shows a visible off-diagonal block at residues 18–25 vs 45–55 precisely matching the known $\beta$-hairpin contacts between strands $\beta_1$–$\beta_4$. This same off-diagonal block reappears in GB1's perturbation sensitivity map, demonstrating that mutations at long-range contact-forming positions produce the largest embedding displacements - the attractor basin is most strongly defined precisely where the fold is held together! The hydrophobic interaction matrices show block structure that spatially coincides with the brightest contact regions across all three proteins, confirming that ESM-2's attention preferentially tracks hydrophobic contacts [2] rather than all pairwise interactions equally.

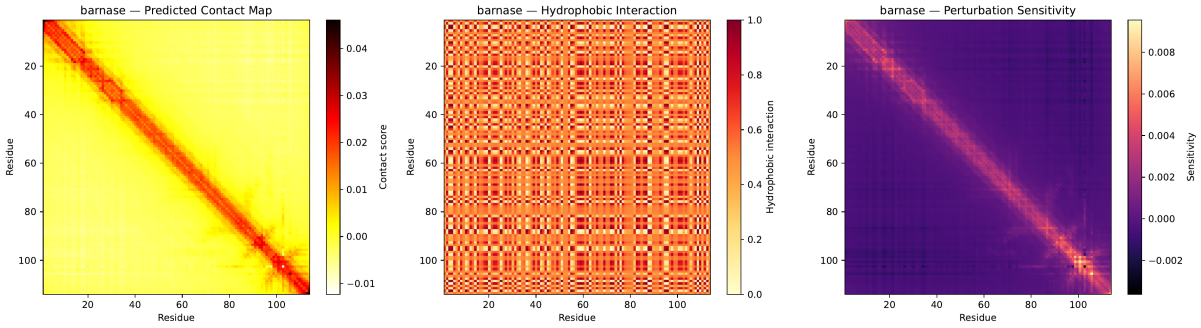



Figure 10: Contact maps - barnase. Left: predicted contact (APC-corrected). Centre: hydrophobic interaction. Right: perturbation sensitivity.

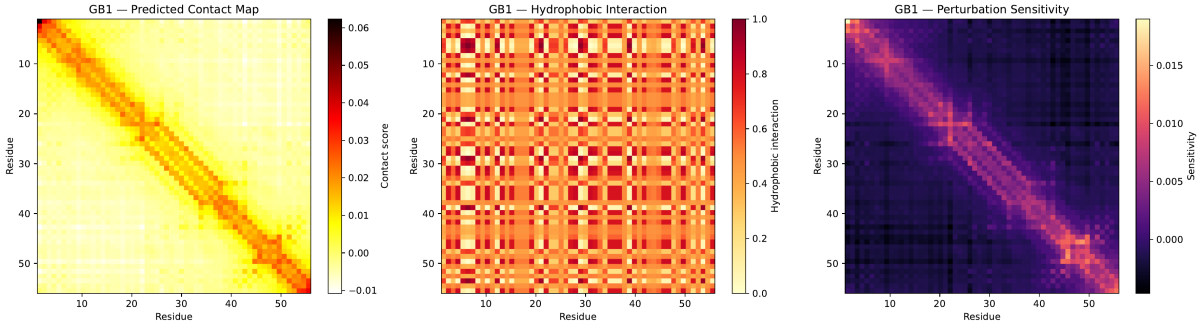



Figure 11: Contact maps - GB1.

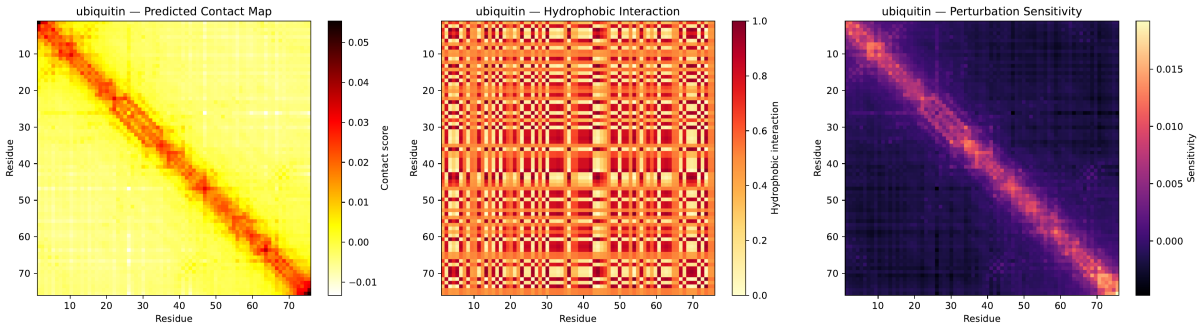



Figure 12: Contact maps - ubiquitin.

## 3.7 Positional Sensitivity and Hydrophobicity

Positional sensitivity plots show the mean L2 displacement per residue position overlaid with Kyte-Doolittle hydrophobicity. Across all three proteins, buried hydrophobic residues tend to show lower sensitivity (smaller mean L2) compared to surface-exposed or loop residues. This pattern is consistent with the attractor basin model: core hydrophobic residues are *buffered* by the attractor well (mutations at these positions are constrained by the strong hydrophobic interactions holding the core together), while surface residues, being less constrained, produce larger embedding perturbations per mutation. GB1 shows asymmetric sensitivity concentrated in the first 30 residues, corresponding to its $\beta$-hairpin and helix region.

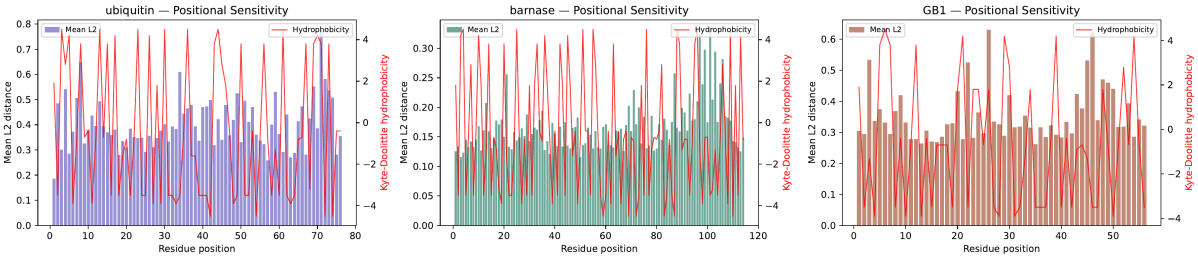



Figure 13: Positional sensitivity (mean L2, bars) with Kyte–Doolittle hydrophobicity overlay (red line) for ubiquitin, barnase, and GB1.

## 3.8 t-SNE Visualisation

t-SNE embeddings at perplexities 15 and 30 were computed from PCA-reduced (50-dimensional) displacement vectors. The protein-coloured panels show partial separation of the three proteins, with barnase most distinct, while ubiquitin and GB1 overlap more, suggesting shared embedding geometry. Colouring by stability reveals a radial gradient too: destabilising mutations (red) concentrate at the periphery, while neutral and stabilising mutations (blue) cluster near the centre! Colouring by L2 distance confirms the same pattern. The consistency of this gradient across two perplexity settings indicates it is a robust feature of the data rather than a t-SNE artefact.

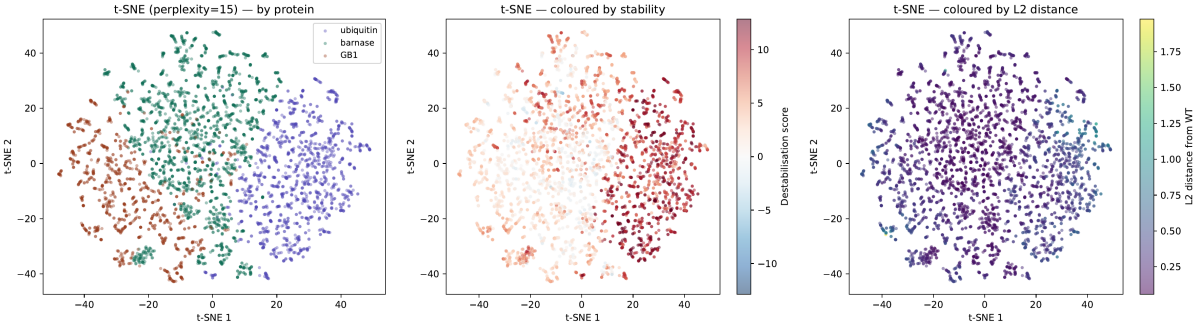



Figure 14: t-SNE (perplexity = 15) coloured by protein identity, stability score, and L2 distance.

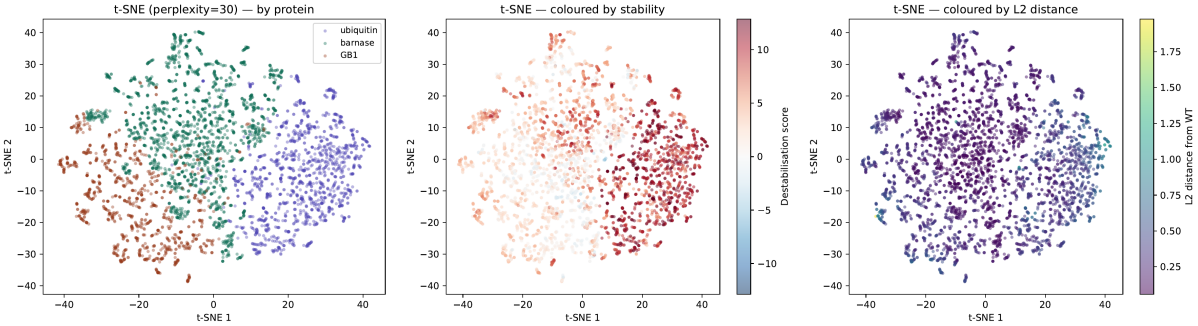



Figure 15: t-SNE (perplexity = 30) coloured by protein identity, stability score, and L2 distance.

## 3.9 UMAP Visualisation

UMAP projections at `n_neighbors` = 10, 15, and 30 corroborate the t-SNE findings with improved global structure preservation. At `n_neighbors` = 10, the three proteins form well-separated clusters with visible sub-structure within each cluster. Higher neighbour counts produce smoother manifolds with more mixing between proteins. Across all settings, the stability and L2 gradients remain clearly visible: destabilising mutants occupy cluster peripheries while stabilising mutants cluster near cluster centres.

The UMAP topology reveals that each protein occupies a distinct region of embedding space, yet the radial stability gradient within each cluster is shared, so it could be a universal attractor-like feature that the pLMs are capable of detecting!

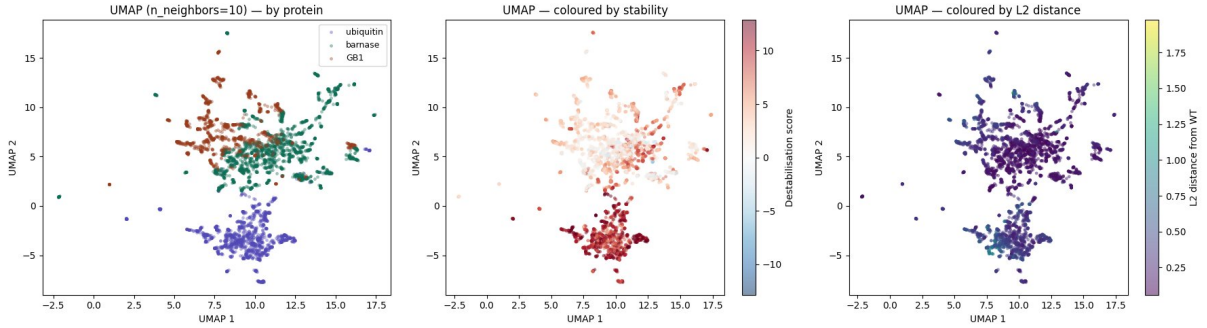



Figure 16: UMAP (`n_neighbors` = 10) coloured by protein, stability, and L2 distance.

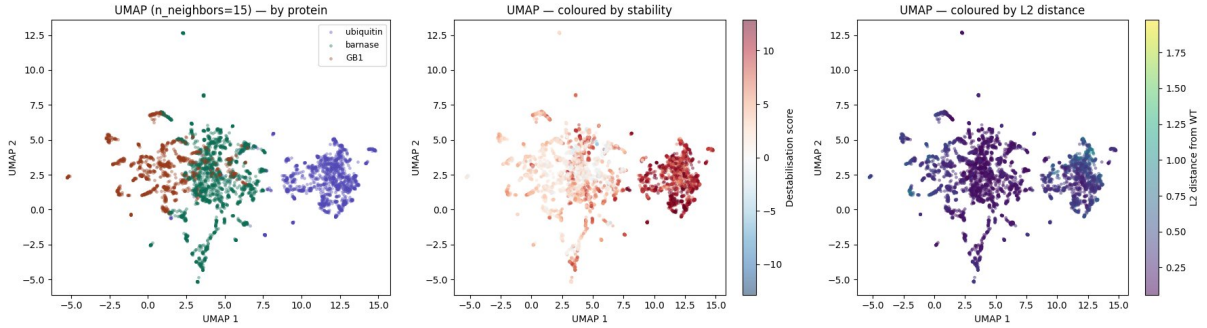



Figure 17: UMAP (`n_neighbors` = 15) coloured by protein, stability, and L2 distance.

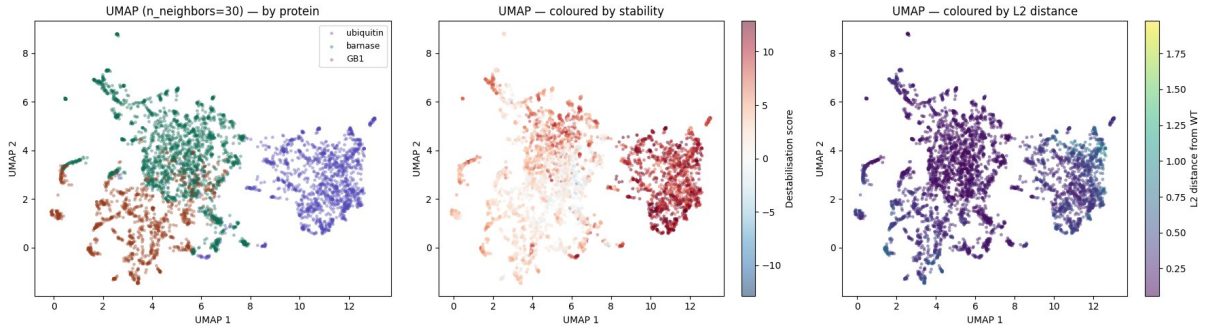



Figure 18: UMAP (`n_neighbors` = 30) coloured by protein, stability, and L2 distance.

# 4 Discussion

## 4.1 Evidence for Attractor Basin Geometry

The convergence of multiple independent analyses provides strong evidence that pLM embedding space exhibits attractor basin geometry around the wild-type sequence. The funnel scores are hundreds of standard deviations above the permutation null, a level of significance that is difficult to attribute to artefact. The PCA projections show exactly the radial pattern predicted by the NLD framework: the wild-type at the origin, surrounded by a cloud of mutants whose distance from the centre correlates with their predicted destabilisation. The anisotropy of displacement vectors adds a further dimension since the attractor has directional structure, with preferred escape directions that are shared among destabilising mutations.

## 4.2 Protein-Specific Differences

GB1 consistently shows the strongest signals across all metrics: highest Spearman $r$ (0.569), highest funnel $z$-score (1,444.5), and best experimental validation ($r = 0.790$). This is consistent with its compact $\beta$-grasp fold, where mutations are tightly constrained by the close-packed hydrophobic core. Barnase, with its larger and more complex $\alpha + \beta$ architecture, shows intermediate coupling. Ubiquitin, despite being well-studied, shows the weakest L2–stability correlation ($r = 0.173$), possibly because its highly stable structure tolerates many mutations without large structural perturbations, compressing the dynamic range of the embedding signal. Nevertheless, even ubiquitin achieves a funnel $z$-score of 75.8, which is still extremely significant.

The barnase validation result has a precise biological interpretation that strengthens rather than weakens the overall argument. The significant within-stabilising-subset anti-correlation ($r = 0.202$, $p = 0.018$) demonstrates that ESM-1v's proxy is measuring evolutionary sequence plausibility, not thermodynamic free energy directly. These two quantities are correlated across the range of naturally occurring variation, which is why the proxy works for GB1 and trends correctly for ubiquitin, but diverge when mutations are engineered into stability regimes that natural selection has not explored. The barnase S2648 entries are precisely such a set: they were identified through directed engineering experiments, not evolutionary sampling.

This finding has a broader implication for the NLD framework: the attractor in pLM embedding space is an evolutionary attractor, not a pure thermodynamic one. The native state occupies the centre of the basin because it is both thermodynamically stable and evolutionarily accessible; over-engineered stabilising mutants sit at the periphery not because they are far from the thermodynamic minimum but because they are far from the evolutionary optimum. The two attractors are aligned for most natural proteins and most mutations, but barnase exposes the seam between them.

### 4.3 Biological Interpretation of Contact Maps

The attention based contact maps provide an interpretability window into what ESM-2 has learned. The predicted contacts recover known secondary-structure features (the $\alpha$-helix/$\beta$-sheet topology of barnase, the $\beta$-hairpin of GB1), and the perturbation sensitivity maps show that positions at structural contacts are also those where mutations produce the largest embedding displacement. This coupling between attention structure and perturbation sensitivity supports the conclusion that ESM-2 has acquired genuine biophysical knowledge, not simply sequence statistics. This knowledge gives the embedding space its attractor-like structure.

The hydrophobic interaction matrices further confirm that hydrophobic residue clustering shapes the attractor geometry, consistent with Dill's thermodynamic framework [2].

### 4.4 Mutational Trajectories and Dynamical Complexity

The present analysis treats each single-point mutation as an independent perturbation from the wild-type, which ignores the sequential, path-dependent nature of real life mutational trajectories. Under the NLD framework, compounding mutations correspond to orbits in embedding space whose geometry relative to the attractor encodes cumulative substitution effects. Whether mutational trajectories exhibit signatures of chaotic behaviour, such as exponential divergence of nearby orbits or positive Lyapunov exponents, remains an open question. Future work mapping double and higher-order mutations in ESM-2 embedding space, combined with finite-time Lyapunov exponent estimation, could directly test this hypothesis and provide a dynamical systems interpretation of the well-documented context-dependence of mutational effects (epistasis).

## 5 Limitations

Several limitations should be noted:

- The ESM-1v masked-marginal score is a *proxy* for $\Delta\Delta G$, not a direct measurement; its negative correlation with barnase experimental data highlights dataset-specific biases.
- Our analysis is restricted to single-point mutations and three proteins; extending to multi-site mutations and a broader protein set would strengthen generalisability.
- The pLM embedding space, while a useful proxy, is not a physical folding trajectory: we are testing attractor-like geometry in *representation space*, not in conformational space. The NLD framing is therefore a conceptual analogy grounded in empirical observation, not a literal dynamical model.
- The S669 dataset could not be used for validation due to identifier format mismatch, reducing the independent validation set.

# 6 Conclusion

This project demonstrates that protein language model embedding space, specifically, ESM-2's 1,280-dimensional hidden-state space, exhibits geometric properties consistent with attractor basin topology when probed by systematic single-point mutagenesis. Across three model proteins, we observe:

1. Positive, significant correlations between embedding displacement and predicted destabilisation.
2. Funnel scores 75–1,445 standard deviations above the permutation null.
3. Anisotropic displacement vectors with directional structure.
4. Attention-derived contact maps that recover known structural features and couple to perturbation sensitivity.
5. Consistent radial stability gradients in PCA, t-SNE, and UMAP projections.

These findings support two complementary conclusions. First, the NLD attractor framework provides a useful lens for understanding protein folding landscapes, complementing traditional thermodynamic descriptions. Second, pLMs such as ESM-2 acquire genuine biophysical knowledge during pretraining, their embedding spaces encode structural stability, contact topology, and hydrophobic organisation in a geometrically structured manner that goes well beyond simple sequence statistics.

Some ideas for Future work could include extending this analysis to also have additional proteins with diverse topologies, molecular dynamics trajectories for direct comparison between embedding-space and conformational-space basins, and possibly supervised models trained on the extracted features to predict experimental $\Delta\Delta G$ values.

# 7 Acknowledgments

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